

Factory hall mission — camera navigation with cloud object detection (v2.0.0)

Generated 2026-10-04 from `hall_mission_20261004_031720`. PX4 SITL v1.18 + Gazebo Harmonic + ROS 2 Humble.

Verification level L1–L2: simulation only. These results show what the simulated drone did in a simulated hall. They do not show that the real aircraft is safe to build or fly.

Summary

The drone takes off from a pad, flies a 27.5 m factory aisle with five obstacles that each leave one gap (alternating sides), scans the room at the far end, flies back through the aisle and lands. The only obstacle information it has is what its camera sees: frames go over an emulated factory WiFi link to a cloud object detector, and the boxes that come back are turned into a map on the drone. **Of four clean runs, three completed the whole mission** with no contact, at least 0.52 m of clearance, and landings within 0.09 m of the pad centre. The fourth got stuck on the way back; a dead-end escape was added after it but has not yet been exercised in flight. **Detection recall fails its gate** (0.76–0.79 against 0.90).

Complete missions

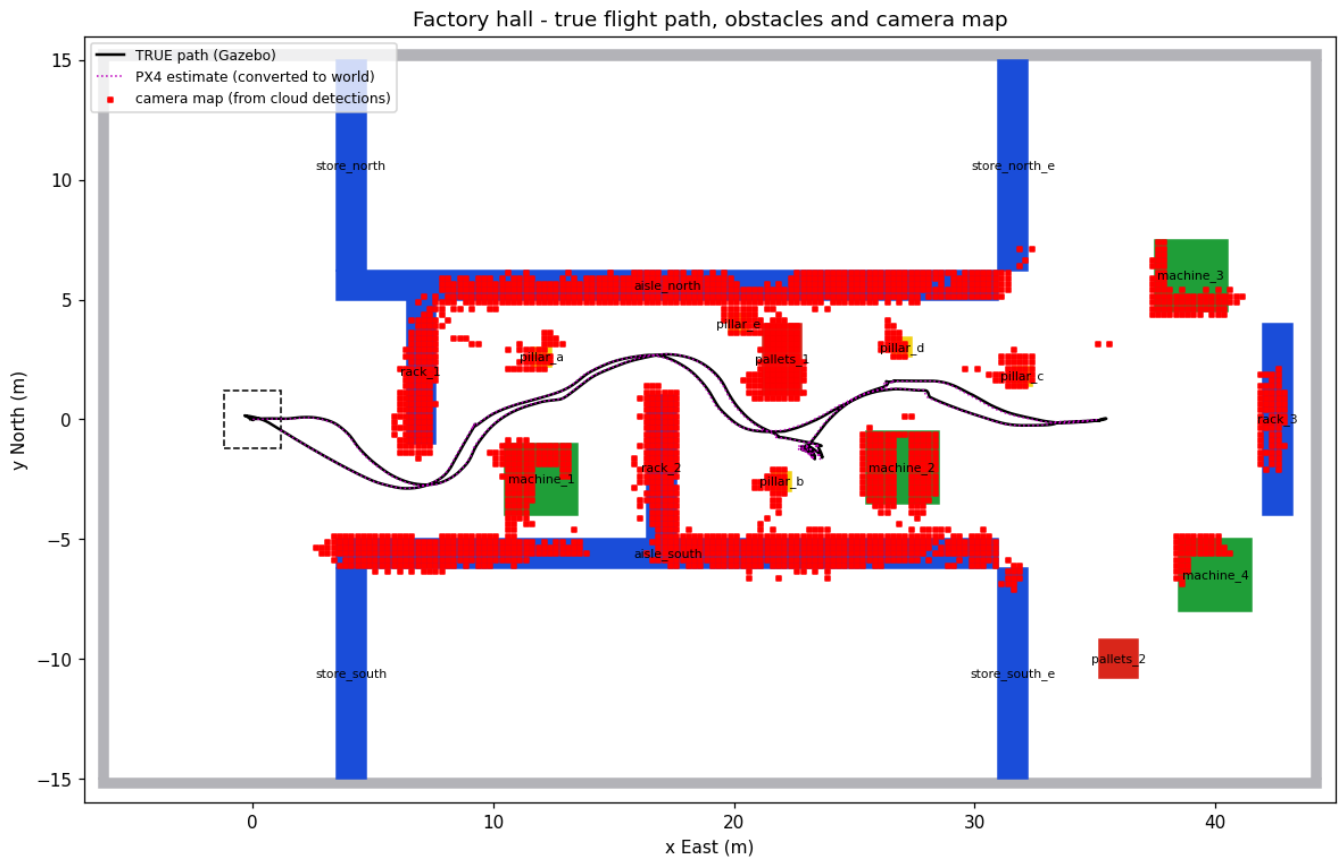
Run	Total	Out / scan / back	Min clearance	Landing error	Recall	Range err. median / p90	Round trip median / p95
11 (headless)	120.6 s	47.5 / 13.0 / 49.8 s	0.523 m	0.089 m	0.758	2.1% / 8.7%	117 / 174 ms
12 (watched)	142.0 s	— / — / 74.6 s	0.521 m	0.083 m	0.787	1.9% / 8.3%	118 / 166 ms
14	stuck on return	51.6 / 12.8 / —	0.556 m	—	0.837	2.9% / 8.6%	122 / 168 ms
hall_mission_20261004_031720 (recorded)	140.2 s	67.7 / 12.0 / 50.5 s	0.540 m	0.077 m	0.786	2.2% / 9.3%	121 / 170 ms

Recorded run status: **complete - returned to pad and landed**. All clearance, contact and landing figures are measured from Gazebo's true pose, not PX4's estimate. Runs 11 and 12 predate per-run output folders; their figures come from the run logs at the time. A third session was excluded because two run harnesses overlapped on one output folder.

Gates (recorded run)

Gate	Threshold	Result	Why this threshold
Hull clearance	≥ 0.40 m every sample	PASS	about one prop radius beyond the 0.35 m airframe radius
Contact	none	PASS	
Landing	≤ 0.50 m from pad centre	PASS	pad half-width less the airframe
Detection recall	≥ 0.90 (in view, ≤ 8 m, ≥50% visible)	FAIL	missing 1 in 10 nearby obstacles is too many
Range error	median ≤ 15%	PASS	keeps the 1.2 m stop margin at 7 m look-ahead

True flight path, obstacles and the camera-built map



Black: the drone's true path from Gazebo. Red: cells the drone mapped from cloud detections. The aisle is the only route from the pad (left) to the scan zone (right).

How it works

Camera to cloud. The Arducam (640x360, 15 fps) is read on the drone, JPEG-encoded at 10 Hz (~5–7 KB per frame) and sent over an emulated link: 11 Mbps, 35 ± 15 ms each way, 1% loss. The link is an assumption, not a measured network; drone and cloud both run on one laptop.

Cloud detection. Classical computer vision: colour segmentation per object class, boxes, class labels, confidence, and the pixels where each object meets the floor. Chosen because the hall's objects are synthetic coloured shapes a pretrained detector would not recognise.

On the drone. Each floor-contact pixel, with the camera height and the IMU attitude from PX4's EKF at capture time, gives a range and bearing. These build an occupancy map (0.25 m cells). Free space seen along each ray clears stale cells. A* plans over the map; floor never seen is passable but costs more. Local avoidance checks the next 2.5 m.

Safety stays on the drone. It holds position if detections are older than 0.5 s (link lag or outage), and stops if anything is within 1.2 m ahead. It never depends on the cloud for safety.

Positioning is PX4's EKF with simulated GPS (team decision for this version). Indoors there is no GPS; camera + IMU localisation is the next major step.

Two bugs that invalidated earlier results

Bug	Effect	Fix
Waypoints in Gazebo's frame (East-North-Up) sent to PX4 unconverted (it reads North-East-Down)	Every v1 mission flew the course rotated 90°, beside the walls. All checks used the same wrong frame, so nothing flagged it.	One conversion module with unit tests; all judging done against Gazebo's true pose.
Magnetic declination mismatch (world at California coordinates)	PX4 heading 26° wrong on the pad; camera detections rotated on the map.	PX4's stock world origin; heading error now −4.8° on the pad, ~0–5° in flight.

Retracted

All v1 factory-mission results: gates threaded, zero collisions, 71–74% camera-steered, 91–92 s mission time, the clearance figures and the camera+IMU fusion numbers. The v1 course was flown rotated 90°. This also explains the unresolved v1 contradiction (0.00 m clearance with no collision). Hover physics, thrust, power and endurance results are not affected. Also retracted: the claim that cut-off boxes give an upper bound on range (disproved in 45 of 70 cases).

Open items

- Detection recall below gate (0.76–0.79 vs 0.90); shelf racks weakest (0.62–0.67). Suspected cause: adjacent same-colour racks merging into one box. Not yet confirmed.
- Too few runs to quote a spread with confidence (plan: $N \geq 5$).
- Scripted link-outage test not yet run.
- Residual heading error ~0–5° (≈ 0.6 m at 7 m).
- Positioning is simulated GPS; camera + IMU localisation not yet ported to the hall.

Reproduce

bash ~/drone_sim/scripts/hall_run.sh (headless, judged against truth) · MODE=frame_check bash
~/drone_sim/scripts/hall_run.sh (position and heading check) · scripts/run_sim.ps1 from Windows to
watch with windows. Full write-up: docs/10-factory-hall-v2.md.